

Random Experiment:

Total possible outcomes are known but the occurrence of an outcome is uncertain
e.g.: Tossing a coin,

Probability: is the measure of uncertainty of such a random exp.

$$\text{prob} = \frac{\text{No. of fav. outcomes}}{\text{Total no. of outcomes}}$$

Sample Space (S)

Event: $E \subseteq S$

Types of Events:

Certain Event : $P(E) = 1$

Impossible Event : $P(E) = 0$

Mutually Exclusive : $E \cap F = \emptyset$

Exhaustive Event : $E \cup F = S$

Ex: $S = \{1, 2, 3, \dots, 100\}$

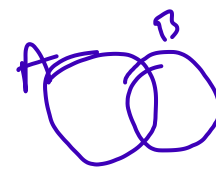
$E_1 =$ Event of getting even no.
 $= \{2, 4, \dots, 100\}$

$E_2 =$ Event of getting odd no.
 $= \{1, 3, \dots, 99\}$

Addition Rule: A & B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = ?$$



$$\frac{n(A \cup B)}{n(S)} = \frac{\frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \cap B)}{n(S)}}{1}$$

Conditional Probability:

Two
events

A & B

(S)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

, $P(B) \neq 0$

$$P(A|A) = 1$$

$$P(S|A) = 1$$

↳ multiplication theorem:
 $P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$

Independent Events:

$$P(E \cap F) = P(E) \cdot P(F)$$

$$\text{or } P(E|F) = P(E), \quad P(F|E) = P(F)$$

Partition of a Sample Space:

$\{E_1, E_2, \dots, E_n\}$ is said to be a partition of a sample space S if

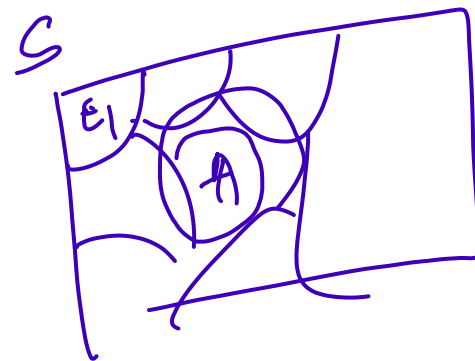
(i) E_i 's are exhaustive events $E_1 \cup E_2 \cup \dots \cup E_n = S$

(ii) E_i 's are mutually exclusive $E_i \cap E_j = \emptyset \quad \forall i \neq j$

Total probability :

$\{E_1, E_2, \dots, E_n\}$

S



$A \subset S$

$$P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + \dots + P(E_n)P(A|E_n)$$

$$= \sum_{i=1}^n P(E_i)P(A|E_i)$$

$$P(A) = P(A \cap S)$$

$$= P(A \cap (E_1 \cup E_2 \cup \dots \cup E_n))$$

$$= P(A \cap E_1) \cup P(A \cap E_2) \cup \dots \cup P(A \cap E_n)$$

$$= \sum_{i=1}^n P(A \cap E_i)$$

$$= \sum_{i=1}^n P(E_i)P(A|E_i)$$

Bayes theorem?

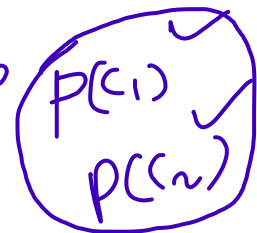
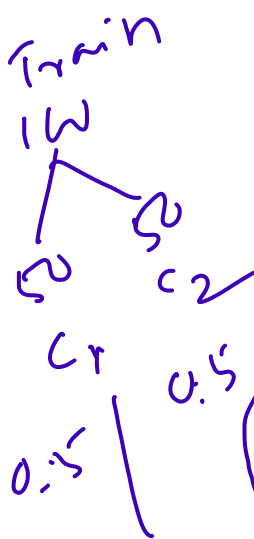
x $P(x)$

$z = 1, 2, \dots, n$
C

1

$$P(E_i | A) = \frac{P(E_i) P(A | E_i)}{\sum_{j=1}^n P(E_j) P(A | E_j)}$$

Classification
Task



$x \in C_1$
 C_2

Decision Rule

$$\frac{P(E_i | A)}{P(A)} = \frac{P(E_i) P(A | E_i)}{P(A)}$$

posterior prob.
prior prob.
class conditional probs.
Bayes

$P(C_1) P(x | C_1)$
 $P(C_2) P(x | C_2)$

$P(C_1 | x) = \frac{P(C_1) P(x | C_1)}{P(A)}$

$x \in C_1$
 $x \in C_2$

test sample

$P(C_1) P(x | C_1)$
 $P(C_2) P(x | C_2)$

C_1 & C_2

ignored

$\frac{P(C_2) P(x | C_2)}{P(A)}$

Random Variable:

$$S = \{HH, HT, TH, TT\}$$

X : No. of heads

$$\begin{matrix} X: & 0 & 1 & 2 \\ p(x): & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{matrix} \rightarrow \textcircled{1}$$

$$p(x) \geq 0$$

$$\sum_2 p(x) = 1$$

$$i=1,2,3$$

$$p(x_i) = P(X=x_i)$$

$$p(0) = P(X=0)$$
$$p(1) = P(X=1)$$
$$= \frac{1}{2}$$

- Discrete RV (Probability Mass function)
- Continuous RV (density ")

$$f(x) \geq 0$$
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Expectation of a RV:

$$E[X] = \begin{cases} \sum x p(x) \\ \int_{-\infty}^{\infty} x f(x) dx \end{cases}$$

↓
mean of X

for D RV

for C RV

Entropy

$$* E[g(x)] = \begin{cases} \sum g(x) p(x) \\ \int_{-\infty}^{\infty} g(x) p(x) dx \end{cases}$$

$$E[x^2] = \sum x^2 p(x)$$

Proposition:

$$E[ax+b] = E[ax] + E[b] \\ = a E[x] + b$$

$$\begin{aligned} &\downarrow \\ &= \sum (ax+b) p(x) \\ &= \sum (ax p(x) + b p(x)) \\ &= a \sum x p(x) + b \sum p(x) \\ &= a E[x] + b \end{aligned}$$

$$E[X_1 + X_2 + \dots + X_n] = E[X_1] + E[X_2] + \dots + E[X_n]$$

Ex: $f(x) = \frac{x+2}{25}$, $x = 1, 2, 3, 4, 5$

$$E[X] = ?$$

Ex: $f(x) = \begin{cases} x^2 & 0 < x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$

$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x^2 dx + \int_1^2 x(2-x) dx$

pdf

= ?

Variance of a RV:

$$\begin{aligned} V[X] &= E[(X - E[X])^2] \\ &= E[(X - \mu)^2] \\ &= E[X^2] - (E[X])^2 \end{aligned}$$

$E[g(x)]$



$$\begin{aligned} & \leftarrow E[X^2 + \mu^2 - 2\mu X] \\ &= E[X^2] + E[\mu^2] - 2\mu E[X] \\ &= E[X^2] + \mu^2 E[1] - 2\mu^2 \\ &= E[X^2] - \mu^2 \\ &= E[X^2] - (E[X])^2 \end{aligned}$$

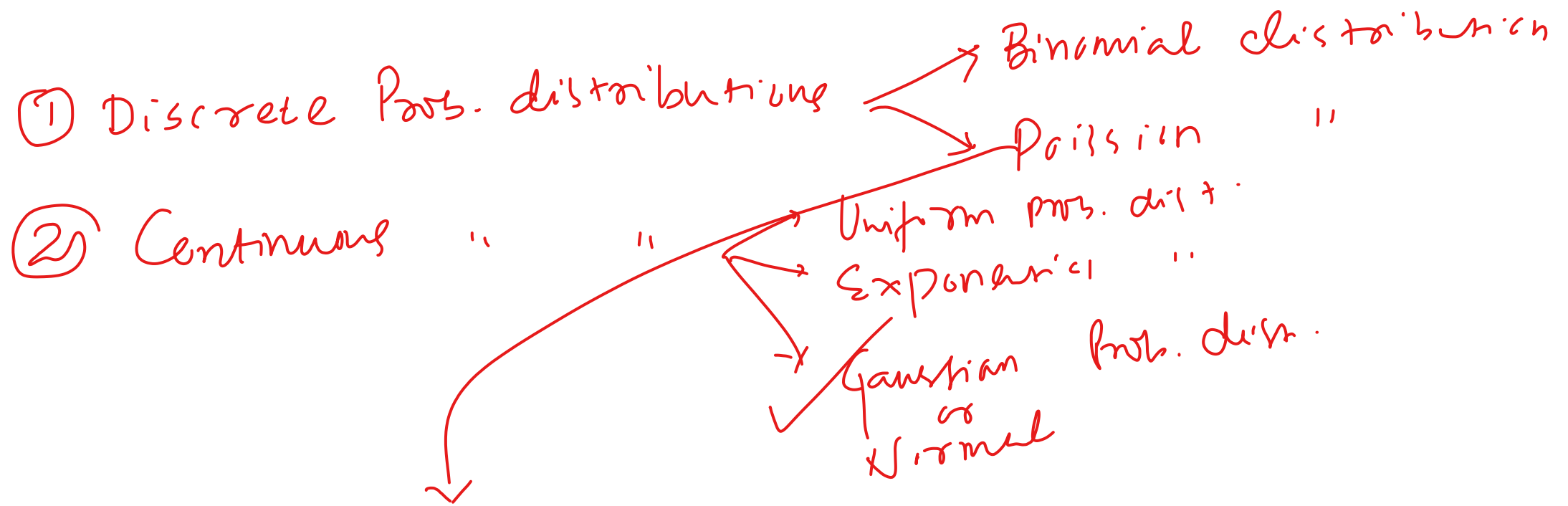
$$\begin{aligned} V(ax+b) \\ &= a^2 V(X) \end{aligned}$$

$$V(aX) = a^2 V(X)$$

$$V(X-a) = V(X)$$

Ex:

pmf
 $f(x) = \frac{x}{15}, x=1,2,\dots,5$
 $E[X]$ & $V[X]$?



$$f(x) = P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$x = 0, 1, \dots, \infty$
 $\lambda > 0$

$$\sum_x P(x) = 1$$

$$\lambda = 2$$

$$\begin{aligned} E(x) &= \lambda \\ V(x) &= \lambda \end{aligned}$$

Uniform

$$f(x) = \begin{cases} \frac{1}{b-a} \\ 0 \end{cases}$$

$$a < x < b$$

otherwise

$$E[X] = \frac{a+b}{2}$$

$$V[X] = \frac{1}{12} (b-a)^2$$

Exponential

$$f(x) = \begin{cases} \frac{1}{\lambda} e^{-\frac{x}{\lambda}} \\ 0 \end{cases}$$

$$x > 0$$

$$x \leq 0$$

$$-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2$$

$$E[X] = \lambda$$

$$V[X] = \lambda^2$$

Gaussian

Univariate
G.D.

$$p(x) =$$

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma}$$

$$\mu =$$

$$\sigma^2 =$$

$$x \rightarrow -3 + 3$$

$$p(x)$$

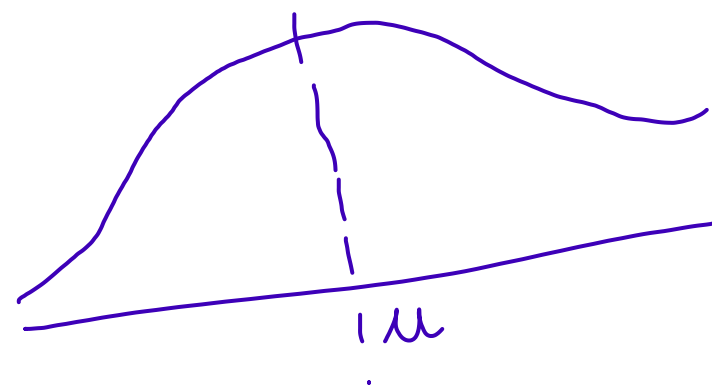
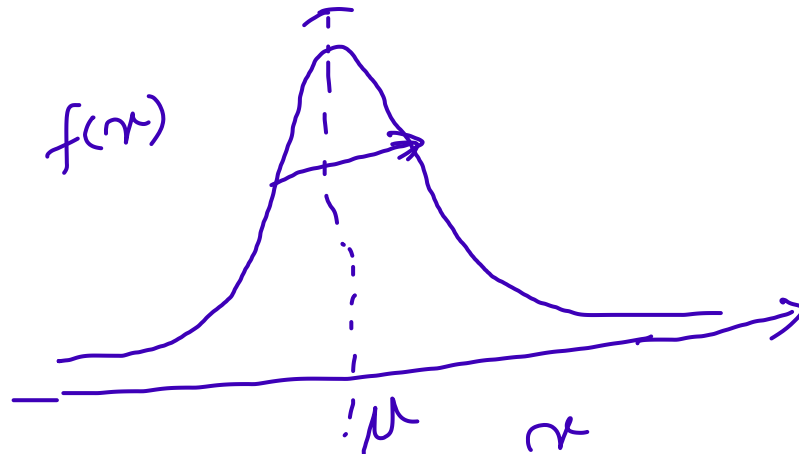
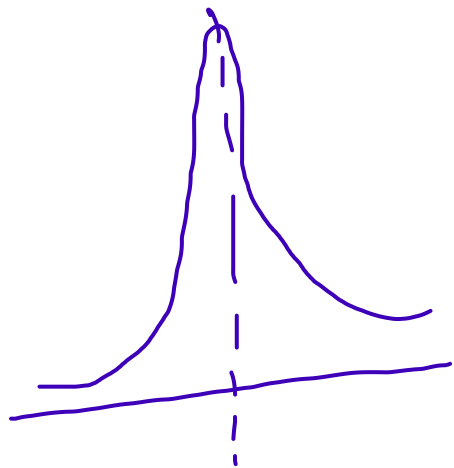
$$\sigma > 0$$

$$-\infty < x < \infty$$

$$-\infty < x < \infty$$

$$E[X] = \mu$$

$$V[X] = \sigma^2$$



- ①
② Central limit theorem

Random Vectors: $x_1, x_2, \dots, x_n \rightarrow n \text{ RVs}$
 $X = [x_1, x_2, \dots, x_n]$ $\hookrightarrow$ Random Vectors

Multivariate Gamma Distribution :